



GAMES OF LEARNING

Name of the qualification:	<i>Diploma in Information Technology in Network Management.</i>
Module Name:	<i>Work Integrated Learning (XINM 5219)</i>
Name of the company:	<i>Games of Learning (GFL)</i>
Group members:	<i>Thato Manyama ST10450306 Siboniso Maruge ST10459440 Andile Rakhombe ST10524430 Winston Dube ST10181381 Sandra Langa st10442678</i>
Slogan:	<i>Gaming minds. Bright Futures.</i>

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1. Executive summary

The background of the project

Games of Learning (GfL) is moving into a new facility and needs a dependable and scalable network to facilitate e-learning, e-books, and online gaming content. This project aims to create and deploy an enhanced network infrastructure that revolves around existing problems and anticipates future expansion.

The objectives of the project

The main primary goal is to establish a reliable and effective network in the new facility, address current network issues, guarantee disaster recovery preparedness, and strategize for growth at both locations. This project's objective is to design and implement a complete IT infrastructure that has workstations, servers, networking, intranet nor internet, website, and office setup.

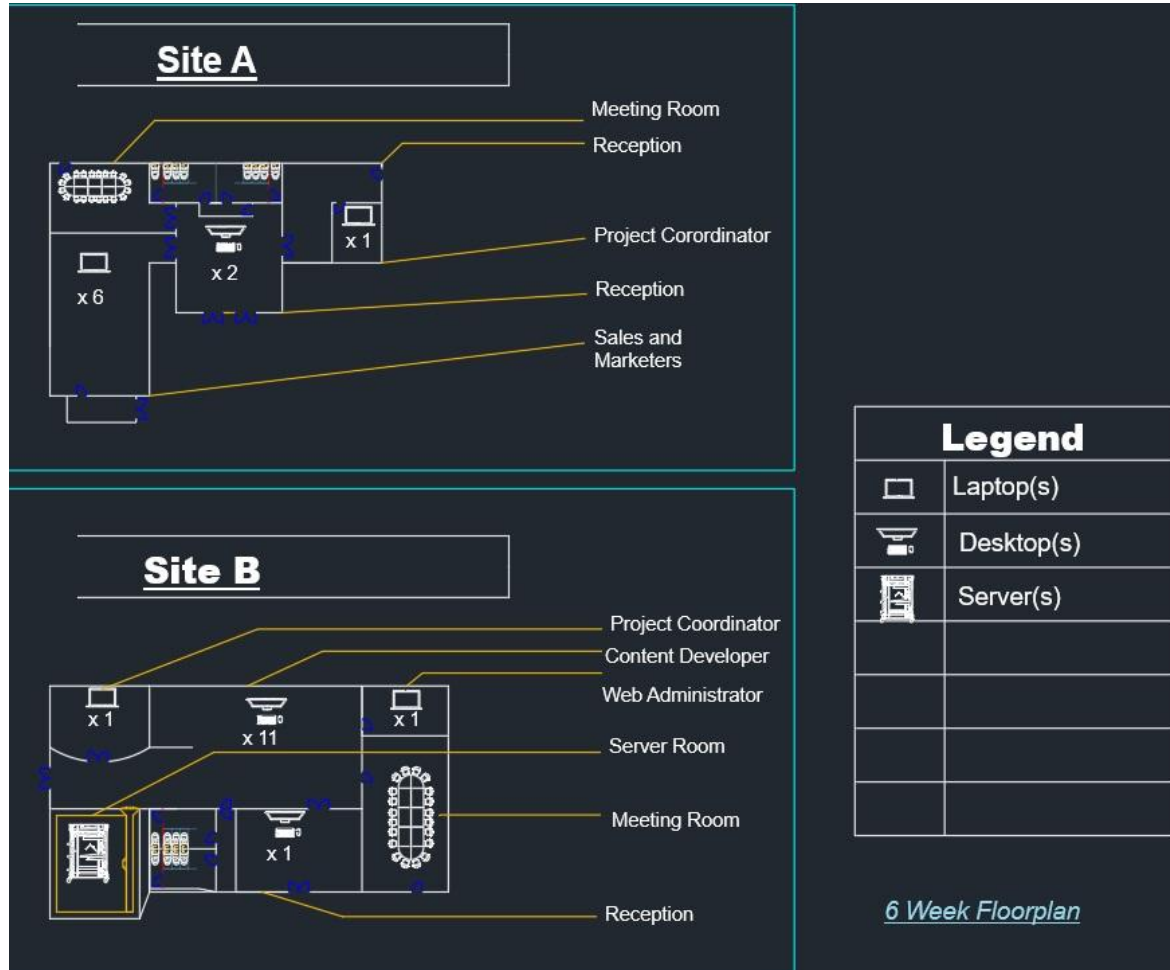
The proposed solutions

The solution here must accommodate a maximum of 23 users in total, including scalability, guaranteed cybersecurity. The suggested approach encompasses creating a new network structure, formulating a comprehensive project schedule (Gantt chart), establishing Standard Operating Procedures, executing a disaster recovery strategy, and setting up a static intranet site for internal purposes.

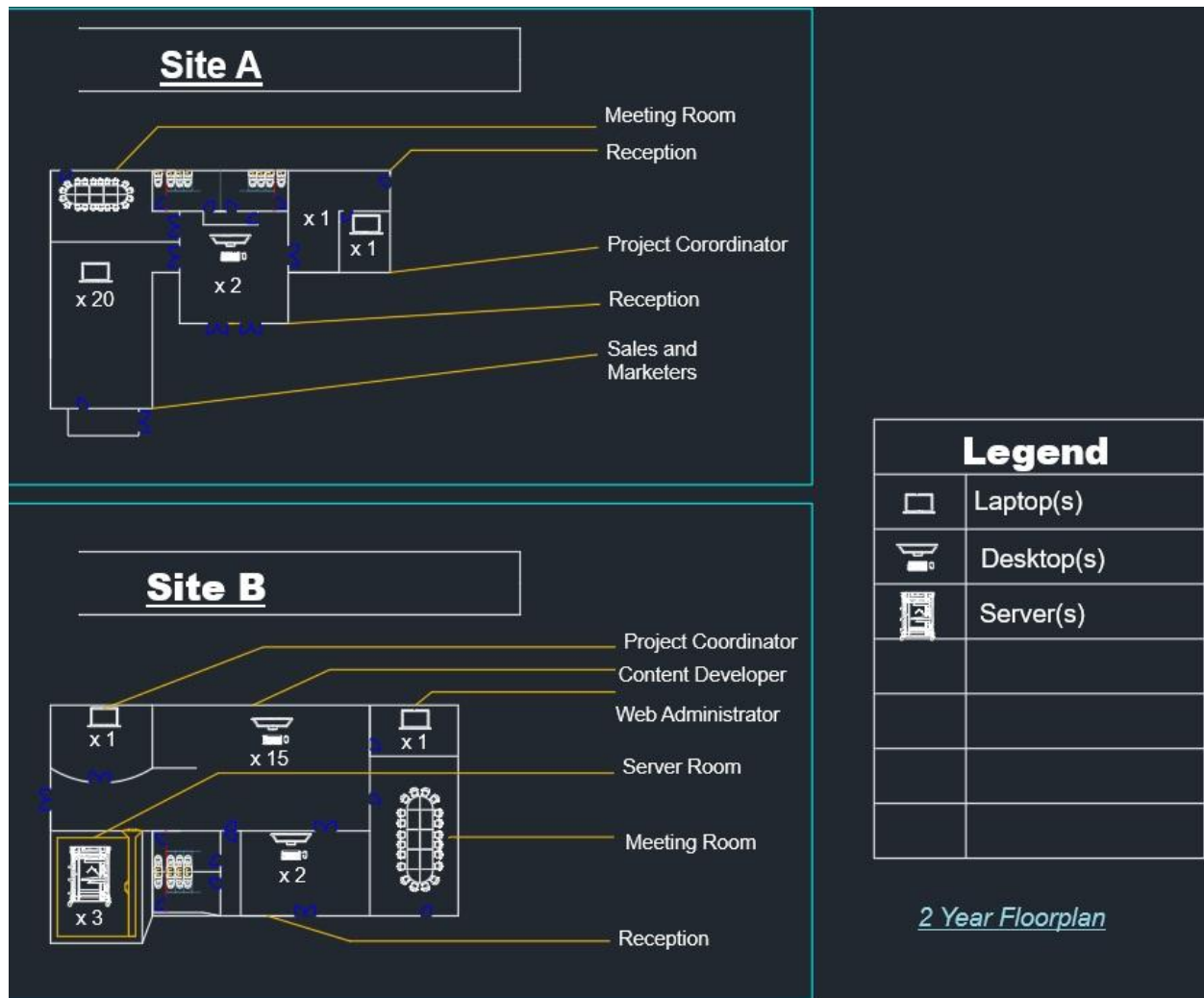
The timeline and budget

This project will be carried out over a period of 6 weeks (about 1 and half months) by a team of 5 individuals. Our estimated budget is R1 564 050, which includes all hardware, furniture, and fire safety systems in case our building catches fire, software, and environmental control measures.

- Site A picture that we will be working on combining it with Site B. The budget for it is R1 564 050.
- Site A and Site B: 6 weeks plan



- **Site A and Site B after 2 years**



2. Milestones and Deliverables

Milestones	Deliverables
Team members names and documentation clarification (IIE, 2025).	The team leader and secretary are appointed. A shared team understanding all the project documents, rules, and responsibilities.
The requirements (IIE, 2025).	A defined analysis identifying the problems with the current IT system (which is the problem domain) and the needs for the new system (which is the solution domain). This includes the lists of the stakeholders, participants, and the current hardware/software.
Document for the project plan (IIE, 2025).	A comprehensive project plan includes: - The defined scope of the new system. - The Task Breakdown Structure (TSB) of all project activities. - A preliminary labor budget based on the TBS. - Risk management assistance with identified risks and mitigation strategies.
Designing networks (IIE, 2025).	A complete design for a scalable network infrastructure that solves the identified problems and is ready for implementation.
Formal recommendations for GfL (IIE, 2025).	A detailed list of the recommendations featuring specific hardware to upgrade, networking issues to rectify, and a created budget to implement these changes.
The Standard Operation Procedures (SOPs) Development (IIE, 2025).	A suite of the SOP documents hosted on a static website includes: - The IT staff structure. - The IT security policy. - The acceptable use policy. - The acquisition procedures - Help desk procedures.
The presentation of the Virtualized Network Solutions (IIE, 2025).	A professional presentation and live stream demonstration. This includes a PowerPoint slideshow, supporting diagrams, a functioning prototype on virtual PC (which is Hyper V), and completed peer evaluation.

3. Work Breakdown Structure

ID	Description of the task	Responsible person	Duration
1	<i>Project planning</i>		5 days
1.1	- Review project documentation & establish team rules	All members	
1.2	- Appointing Project Manager and defining roles	All Members	
1.3	- Analyze case study, identify problems & stakeholders	Sandra Langa	
1.4	- Define project scope & create Task Breakdown Structure (TBS)	Thato Manyama	
1.5	- Develop preliminary labor budget & risk assessment	Thato Manyama	
2	<i>Network system design</i>		10 days
2.1	- Design scalable network architecture to address problems	Andile Rakhombe and Winston Dube	
2.2	- Review and finalize network design with team.	Andile Rakhombe and other team members	
3	<i>Recommendation and Budgeting & inventory sheets</i>		4 days
3.1	- Identify hardware for upgrade and networking issues to fix	Winston Dube and Sboniso Maruge	
3.2	- Develop formal recommendation list and final budget	Thato Manyama	
4	<i>Operational Documentation (SOP)</i>		3 days
4.1	- Draft IT staff structure and all policy documents.	Thato Manyama and Siboniso Maruge	
4.2	- Develop static websites to host all SOPs.	Andile Rakhombe	

5	<i>Disaster Recovery Plan</i>		5 days
5.1	- The purpose of the DRP	Sandra Langa	
5.2	- Risk identification	Sandra Langa	
5.3	- Recovery procedure	Sandra Langa	
6	<i>Presentation preparation</i>		3 days
6.1	- Develop PowerPoint presentation slides	Thato Manyama	
6.2	- Prepare network diagrams and demonstration and system demo	Andile Rakhombe	
6.3	- Rehears presentation and system demo	All members	
7	<i>Final presentation and demonstration</i>		4 days
7.1	- Setup system in venue	Andile Rakhombe	
7.2	- Deliver formal presentation (each member presents their part)	All Members	
7.3	- Answer questions from lecture	All Members	
7.4	- Complete peer evaluation forms	All Members	

4. Risk Management

Every project has potential risks that could delay progress or increase costs, so it's important to plan. For this project, the main risks include people, technical, equipment, economic, and project-related risks.

- a. **People risk:** According to IIE (2025) A possible is that a team member might be absent for a long period or even leave the project. To reduce this risk, we will make sure that every important task has a backup person who can step in if needed (IIE, 2025).
- b. **Equipment risks:** Include things like a server failure or a hard drive crash (IIE, 2025). To manage this, we will have regular data backups, keep spare equipment available, and use a reliable design that allows quick recovery (IIE, 2025).
- c. **Economic risks:** If the project goes over budget, we will include a small contingency amount in the budget and keep a close eye on expenses to avoid overspending (IIE, 2025).
- d. **Technical risks:** Might happen if our team lacks certain skills or if we face compatibility issues with software. To reduce this, we will arrange quick training sessions and test the design carefully before implementation (IIE, 2025).
- e. **Project risk:** Solutions may not be accepted by the stakeholders (IIE, 2025).

4.1. Risk management table

- a. **People risk** (IIE, 2025).

Probability/ Cost	Very low	Low	Medium	High	Very high
Very low					
Low					
Medium					
High				x	
Very high					

- b. **Equipment risk** (IIE, 2025).

Probability/ Cost	Very low	Low	Medium	High	Very high
Very low					
Low					
Medium			x		
High					
Very high					

c. **Economic risk** (IIE, 2025).

Probability/ Cost	Very low	Low	Medium	High	Very high
Very low					
Low					
Medium					
High			x		
Very high					

d. **Technical risk** (IIE, 2025).

Probability/ Cost	Very low	Low	Medium	High	Very high
Very low					
Low					
Medium			x		
High					
Very high					

e. **Project risk** (IIE, 2025).

Probability/ Cost	Very low	Low	Medium	High	Very high
Very low					
Low		x			
Medium					
High					
Very high					

- Feasibility of Study

5.1. Technical Feasibility

According to Martins (2025) Before moving forward with the GfL network project, it is important to confirm that the solution can actually be implemented with the technology, skills, and resources available. This step helps avoid delays and unexpected technical challenges later in the project (Martins, 2025).

For this project, we first looked at the **tools and software** we will need (Martins, 2025). The team will be using Virtual PC to create a simulated network environment where we can safely design and test the new network before installing it in the real building (Martins, 2025). We will also use Microsoft Word to create the required documentation, Microsoft PowerPoint to prepare the final presentation, and other standard software already available in the organization (Martins, 2025). These tools are familiar to the team, and where there are any gaps in knowledge, we can arrange short training sessions to get everyone up to speed quickly (Martins, 2025).

Next, we considered the **hardware requirements**, including network switches, routers, cabling, servers, and workstations (Martins, 2025). These components can all be sourced within the project's budget and are compatible with the company's existing systems (Martins, 2025). The proposed network design is realistic for the size of the new building and is flexible enough to handle future growth, which means it is not only possible to implement but will also continue to work as the company expands (Martins, 2025).

Finally, we looked at the **technical skills** within the team (Martins, 2025). The team members have enough knowledge of networking, server setup, and basic troubleshooting to complete the tasks. Any specialized tasks, such as advanced configuration, can be supported by additional resources or expert advice if required (Martins, 2025).

According to Martins (2025) Based on this analysis, the project is technically feasible. The tools, equipment, and skills required are available, and the network design can be implemented successfully within the given constraints (Martins, 2025).

5.2. Economic Feasibility

Our estimated overall budget= R 1 564 050

According to Vaia (2024) After confirming that the project is technically possible, it is just as important to check whether it is financially realistic. Economic feasibility looks at whether the project can be completed within the available budget and whether the benefits of the project justify the cost (Vaiva, 2024).

For the GfL network project, we considered all the major cost areas, including network hardware (switches, routers, cabling), servers, software licenses, and labor costs for design, configuration, and testing (Vaiva, 2024). A preliminary budget was prepared based on current market prices, and a small contingency amount was added to cover unexpected expenses (Vaiva, 2024). This ensures that if a component fails or a replacement is needed, the project will not be delayed (Vaiva, 2024).

In terms of **cost vs. benefit**, the new network will improve the company’s operations by reducing downtime, increasing speed, and making it easier to scale when new employees or devices are added (Vaiva, 2024). These improvements will save time, reduce technical issues, and support the company’s goal of expanding its e-learning services. This means the money spent on the project is likely to be recovered over time through better productivity and fewer disruptions (Vaiva, 2024).

Since the estimated costs fit within the budget that GfL has set aside for this expansion, and the benefits of the new network are clear and measurable, the project is considered economically feasible (Vaiva, 2024).

- Expanded Budget Breakdown

Computer & Laptops

Category	Item	Quantity	Unit Price (ZAR)	Total (ZAR)
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Computers & Laptops	Developer Desktops (High Spec)	9	R21 500.00	R193 500.00
	General Use Desktops	4	R15 000.00	R60 000.00
	Admin/Management Laptops	6	R20 000.00	R120 000.00
	Docking stations for laptops	5	R19 000.00	R95 000.00
Computers and Laptops subtotal				R464 000.00

Servers & Networking

Category	Item	Quantity	Unit Price (ZAR)	Total (ZAR)
Servers & Networking	NVR Server	1	R 25 000.00	R25 000.00
	Firewall Appliance	1	R50 000.00	R50 000.00
	NAS Backup	1	R30 000.00	R30 000.00
	File & Storage Server	1	R55 000.00	R55 000.00
	Intranet/Web server	1	R45 000.00	R45 000.00
	48-Port Managed Switches	1	R39 000.00	R39 000.00
	Router (Enterprise)	1	R30 000.00	R30 000.00
	UPS Systems (3kVA)	3	R18 000.00	R54 000.00
	Patch Panels & cabling	1	R20 000.00	R20 000.00
	Wireless Access Points	3	R6 000.00	R18 000.00
	Network Racks	2	R12 000.00	R24 000.00
Servers & Networking subtotal				R390 000.00

Peripherals & Devices

Category	Item	Quantity	Unit Price (ZAR)	Total (ZAR)
Peripherals & Devices	Access Switch 24-Port	1	R20 000.00	R20 000.00
	Laser Printers (B/W)	3	R5 000.00	R15 000.00
	Multifunction Printer (Color)	1	R24 500.00	R24 500.00
	Projectors	3	R12 500.00	R37 500.00
	Large Display Screens	2	R30 990.00	R61 980.00

	VoIP Phones	15	R1 940.00	R29 100.00
	CCTV Cameras + NVR Kit	12	R4 000.00	R48 000.00
	Headsets	20	R1 900.00	R38 000.00
	External Backup Drives (8TB)	2	R8 000.00	R16 000.00
Peripherals & Devices subtotal				R290 050.00

Software & Licensing

Category	Item	Quantity	Unit Price (ZAR)	Total (ZAR)
Software & Licensing	Windows Server 2022 Std Licenses	3	R13 000.00	R39 000.00
	Windows 11 Pro Licenses	41	R3 000.00	R123 000.00
	Microsoft 365 Business Std (per us)	41	R2 400.00	R98 400.00
	Antivirus/ERD (per device/year)	50	R810.00	R40 500.00
	Helpdesk Ticketing Software	1	R20 000.00	R20 000.00
Software & Licensing subtotal				R320 000.00

Connectivity & ISP (Annual)

Category	Item	Quantity	Unit Price (ZAR)	Total (ZAR)
Connectivity & ISP (Annual)	Telkom Business Fiber (200Mbps)	1	R60 000.00	R60 000.00
	Redundant LTE Backup	1	R40 000.00	R40 000.00
Connectivity & ISP (Annual) subtotal				R100 000.00

Grand Total: R1 564 050

5.3. Operational Feasibility

According to IIE (2025), operational feasibility determines whether a project can function effectively in a real-life business environment and whether the organization can support it long term.

For Games of Learning (GfL), the proposed network infrastructure is operationally feasible because it aligns with the company's goals of supporting e-learning and gaming content (Martins, 2025). The network will improve communication between staff and students, enable faster data sharing, and allow remote access to educational resources (Martins, 2025).

Training sessions will be provided to staff and students to ensure they understand how to use the new systems (Martins, 2025). The IT support team will receive additional training in basic network management and cybersecurity practices so they can maintain the system efficiently (Martins, 2025). Routine maintenance plans and Standard Operating Procedures (SOPs) have been developed to ensure continuous operations with minimal disruption (IIE, 2025).

Therefore, from an operational perspective, the system can be adopted smoothly and supported without placing unnecessary strain on existing staff or resources (Martins, 2025).

5.4. Schedule Feasibility

According to Martins (2025), schedule feasibility checks if the project can be completed within the time available using the resources and team assigned.

The GfL project is planned for six weeks, which includes planning, design, implementation, testing, and presentation (Martins, 2025). Based on the Gantt chart and Work Breakdown Structure, tasks are divided efficiently among the five team members (Martins, 2025). Each milestone has clear dependencies, and parallel tasks (such as hardware procurement and documentation preparation) allow time to be used effectively (Martins, 2025).

Buffer days have been included for unforeseen delays such as equipment delivery or software installation issues (Martins, 2025). Since the tasks fit comfortably within the six-week period and resources are adequate, the project is considered schedule-feasible (Martins, 2025).

5.5. Legal Feasibility

Legal and organizational feasibility focuses on whether the project complies with regulations and the company's internal policies (Vaia, 2024).

All software used in the GfL network project will be properly licensed to avoid violating copyright laws (Vaia, 2024). The organization will also follow data protection and privacy policies to safeguard student and employee information (Vaia, 2024). User accounts and password policies will comply with standard IT security requirements as outlined in the SOPs (Vaia, 2024).

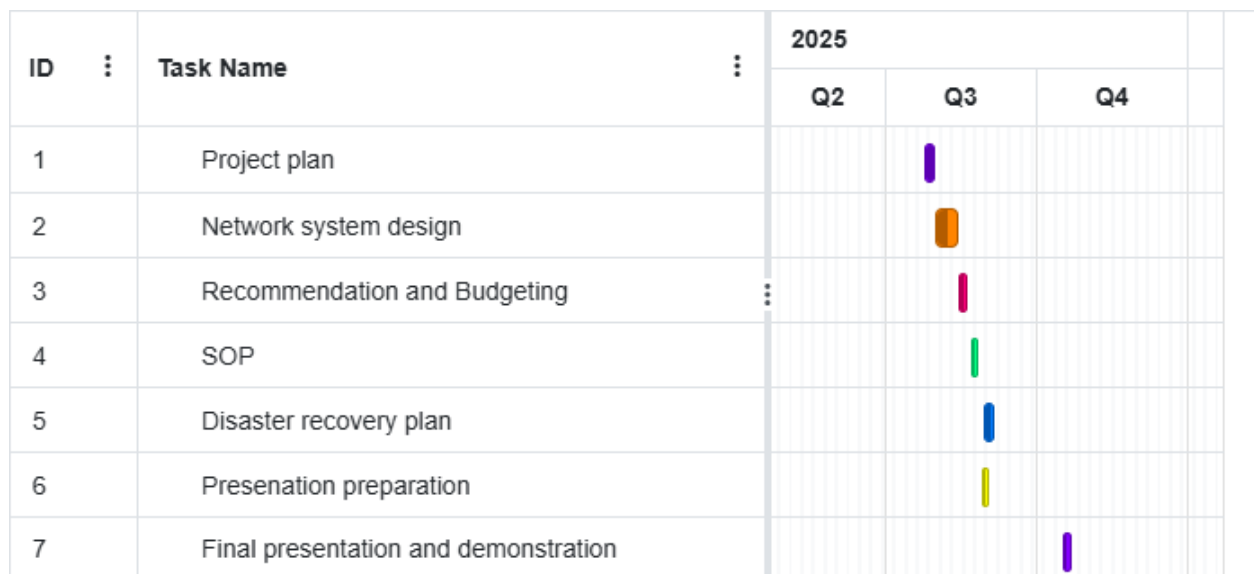
From an organizational point of view, the proposed system fits within the existing structure of the company and supports its strategic plan for digital growth (Vaia, 2024). Therefore, the project is legally and organizationally feasible (Vaia, 2024).

5. Work Breakdown Structure + Gantt Chart and Project Charter

- Work Breakdown Structure

I wrote the work breakdown structure on page 3

- **Gantt Chart**



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(Online gantt, 2025).

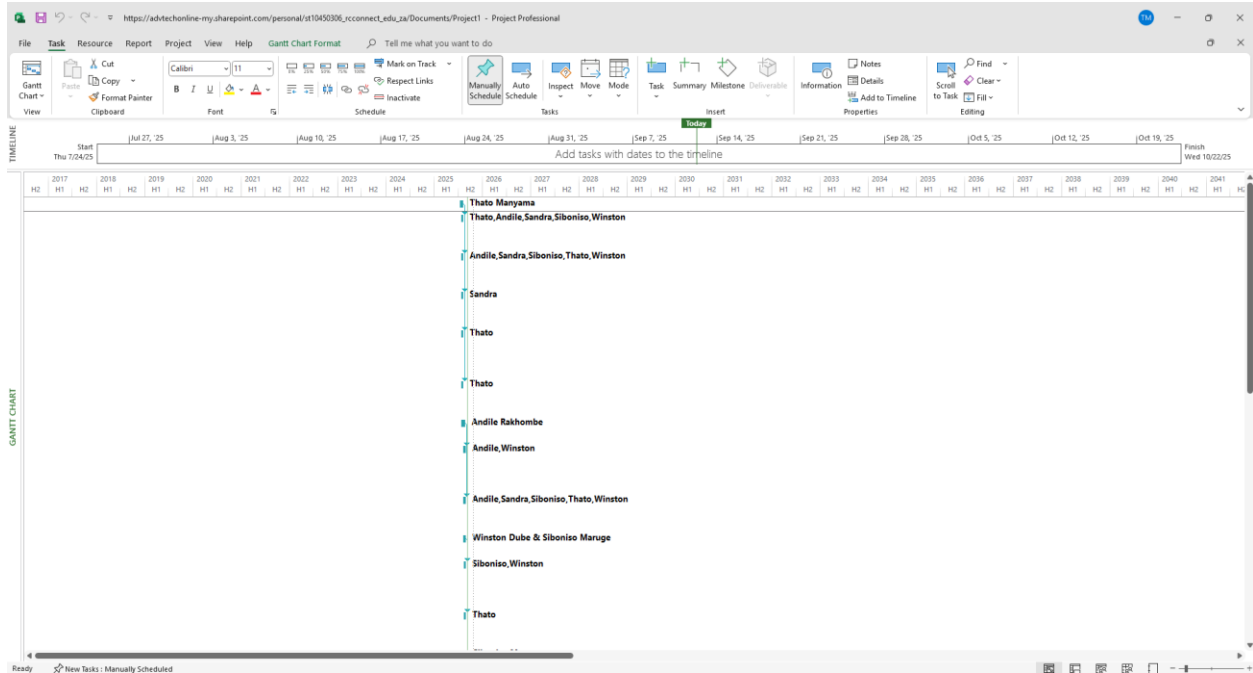
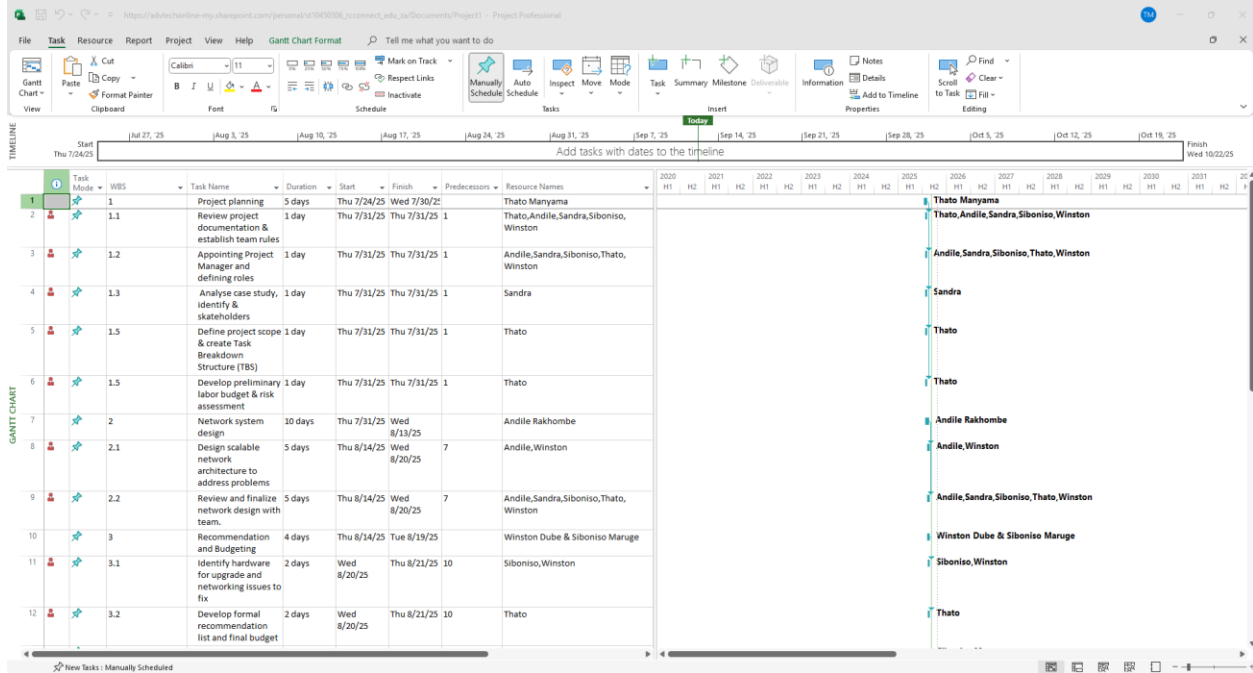
- Project charter

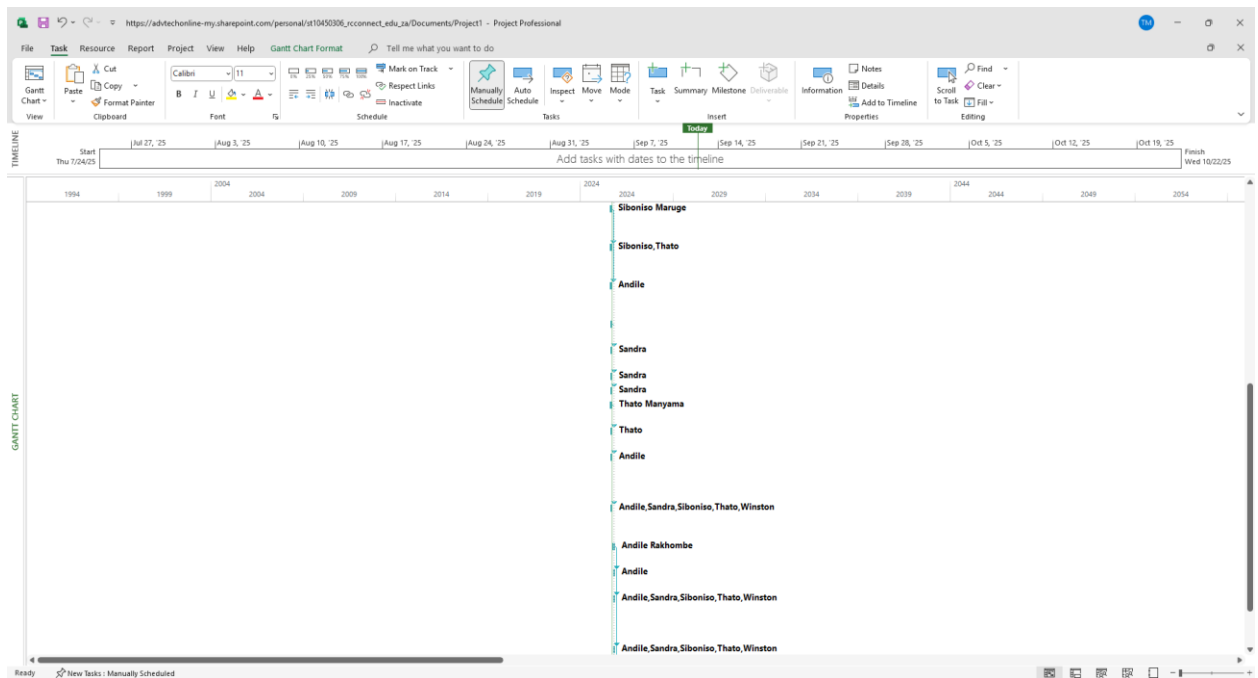
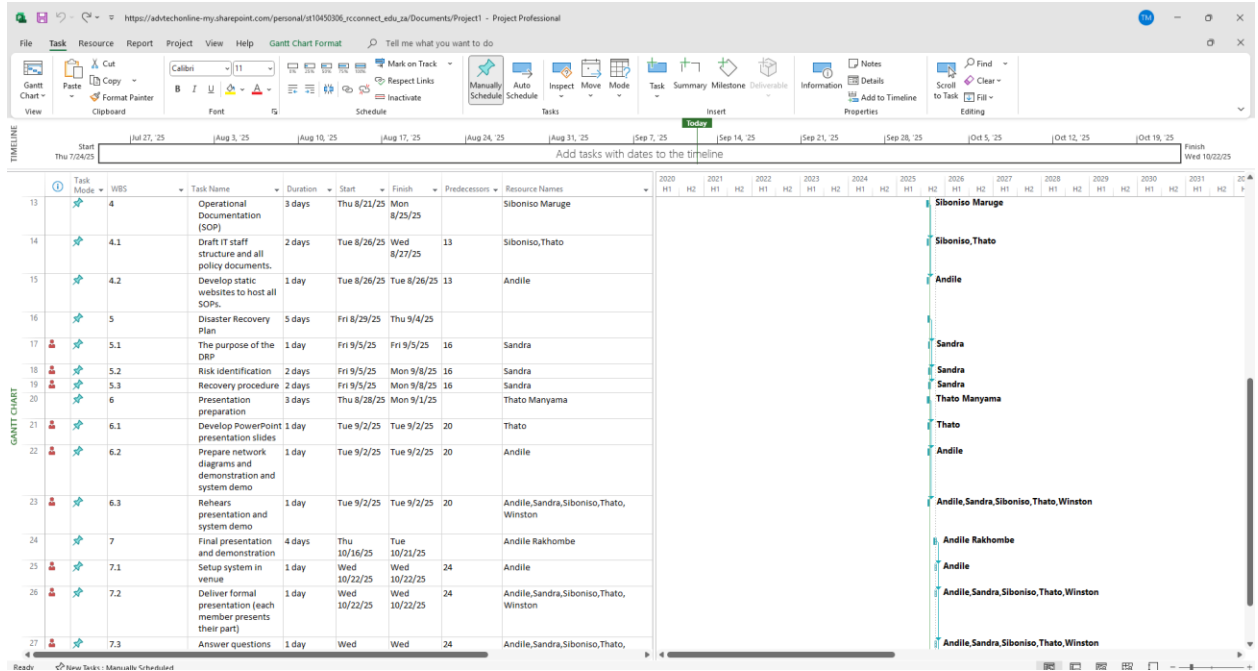
Project Charter

Project Name	GfL Network Expansion and Scalability Project
Start Date	24 July 2025
End Date	4 September 2025
Project Manager	Thato Manyama
Project Sponsor	GfL Management
Project Purpose	To design and implement a new network for the additional building, solve existing network issues, and ensure scalability for future growth.
Objectives	1. Create a functional, scalable network design. 2. Document milestones, deliverables, and risk management plan. 3. Provide SOPs and IT policies for GfL. 4. Deploy a static intranet site to host documentation.
Scope	In-Scope: Network design for both buildings, hardware/software identification, SOP development, disaster recovery plan, and intranet site. Out-of-Scope: Physical hardware purchasing and implementation outside the project design phase.
Stakeholders	GfL Management, IT Department, End Users, Project Team,
Deliverables	Project Plan Document, Risk Management Table, WBS, Gantt Chart, Network Design, SOPs & IT Policies, Intranet Site, Final Presentation
Constraints	Limited budget, project must be completed in allocated timeframe, team must use available tools (Virtual PC, MS Word, MS PowerPoint).
The assumptions	All team members will participate actively, resources are available, and GfL will support proposed recommendations.

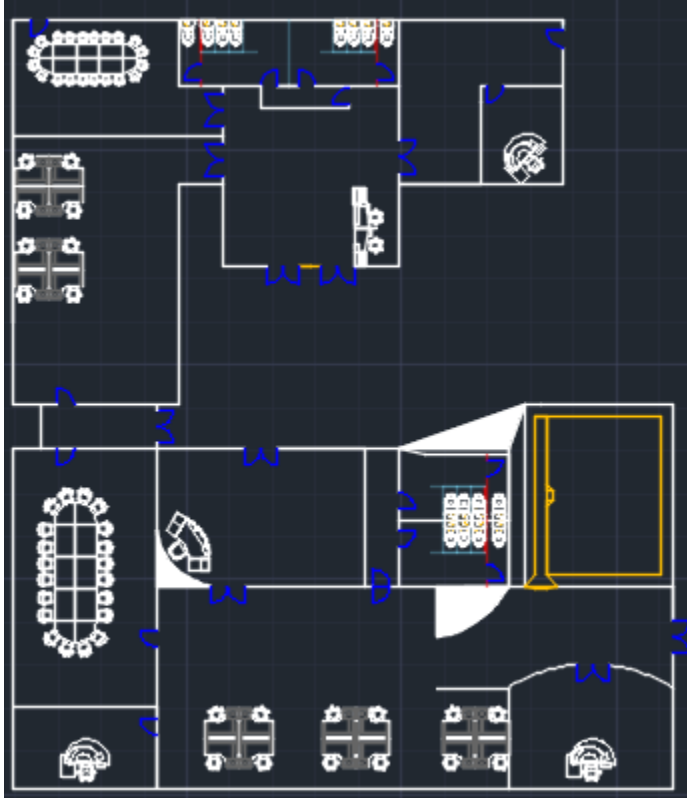
(Canva, 2025)

6. Annexure: Project Report (MS Project)





- The completed floor plan will look this way (Site A + Site B).



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